

Answers

Q	Ans	Q	Ans	Q	Ans
1	b	2	a	3	b
4	c	5	d	6	b
7	b	8	c	9	a
10	b	11	d	12	d
13	a	14	a	15	d
16	d	17	b	18	a
19	b	20	d	21	a
22	b	23	a	24	b
25	c	26	c	27	b
28	c	29	d	30	b
31	c	32	c	33	a
34	c	35	b	36	b
37	c	38	d	39	d
40	a	41	a	42	b
43	b	44	c	45	a
46	d	47	b	48	a
49	b	50	a	51	b
52	a	53	a	54	d
55	b	56	a	57	c
58	b	59	b	60	a

Selected working

(4) Law of multiple proportions: $1.750 \frac{\text{g}}{0.4375} \text{ g} = 4$, so the ratio A : C is 4 : 1.

(26) $N(\text{neutrons}) = A - Z = 27 - 13 = 14$, so $^{27}_{13}\text{Al}$ has 13 protons and 14 neutrons.

(27) $N(\text{neutrons}) = 207 - 82 = 125$, so $^{207}_{82}\text{Pb}$ has 82 protons and 125 neutrons.

(30) $106.90509 \text{ amu} \times 0.5184 + 108.90476 \text{ amu} \times 0.4816 = 55.42 \text{ amu} + 52.45 \text{ amu} = 107.87 \text{ amu}$.

(32) The average, 10.811 amu, lies closer to 11.009 amu (of ^{11}B) than to 10.013 amu (of ^{10}B), so ^{11}B is the more abundant isotope.

(38) $\lambda = \frac{c}{\nu} = \frac{3.00 \times 10^8 \text{ m/s}}{2.4 \times 10^6 \text{ s}^{-1}} = 125 \text{ m}$.

(49) $r = (0.53 \text{ \AA})n^2 = (0.53 \text{ \AA})(2^2) = 2.12 \text{ \AA}$.

(51) $\lambda = \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s} \times 3.00 \times 10^8 \text{ m/s}}{3.03 \times 10^{-19} \text{ J}} = 6.56 \times 10^{-7} \text{ m} = 656 \text{ nm}$.

(54) $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s}}{9.11 \times 10^{-31} \text{ kg} \times 1.00 \times 10^6 \text{ m/s}} = 7.28 \times 10^{-10} \text{ m} \approx 0.73 \text{ nm}$.

(60) Chromium's expected configuration is $[\text{Ar}]4s^23d^4$, but the observed (more stable) configuration is $[\text{Ar}]4s^13d^5$ because a half-filled 3d subshell confers extra stability.